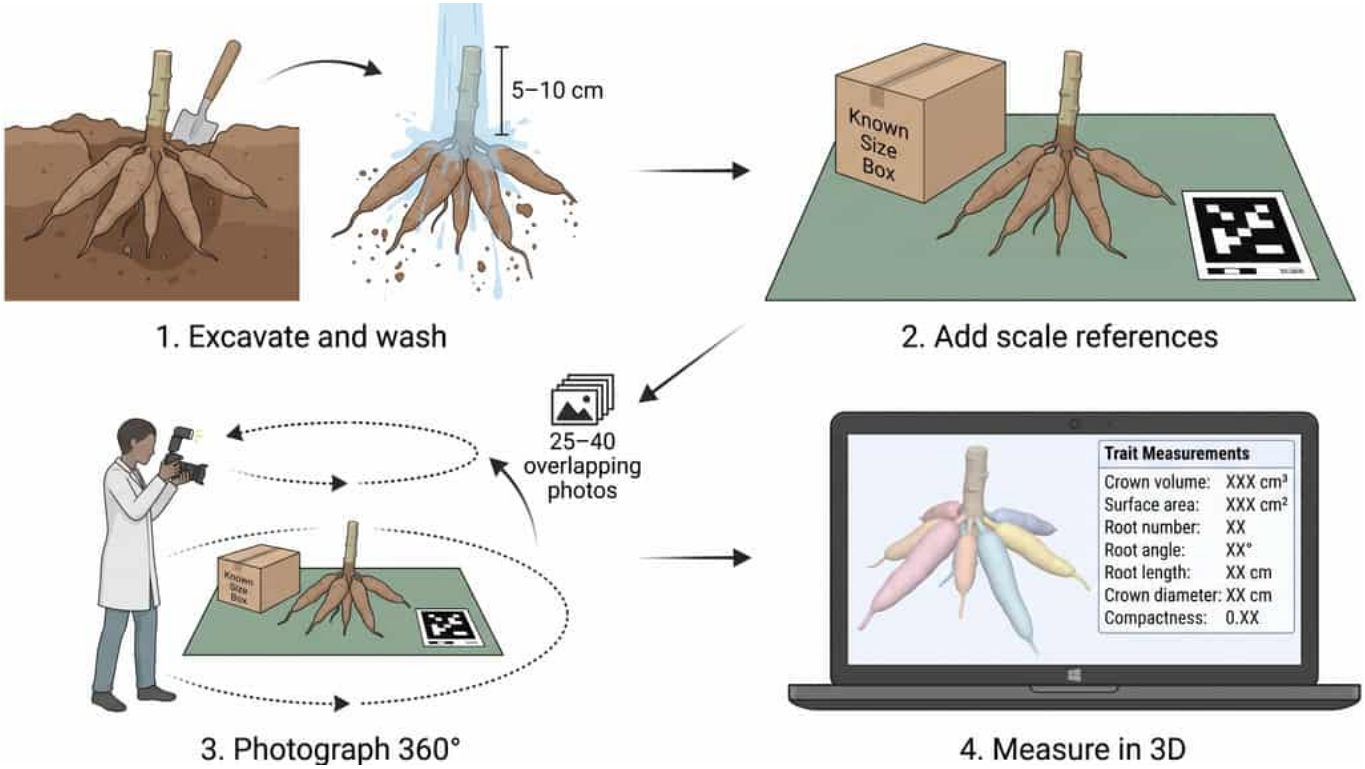
SOP-4 3D root phenotyping

Root crowns (storage roots), washed root systems and CT volumes

SOP-4 · PhenoForge capture SOP · version 1.0 (October 2026)



Method after Sunvittayakul et al. (2022) *Cassava root crown phenotyping using three-dimension (3D) multi-view stereo reconstruction*, Scientific Reports 12, 10030 (doi:10.1038/s41598-022-14325-4). The paper used Agisoft Metashape and Blender; PhenoForge does the same steps with open tools and computes the traits automatically.

Capture a root crown

1 **Excavate and wash**
Dig out the crown, wash off soil, let it drain. Cut the stem leaving a 5–10 cm stub and weigh the roots (fresh weight).

2 **Lay out with scale references**
Lay the crown on its side on a green plastic sheet (the green is removed automatically). Place the ArUco card and, optionally, a box of known size next to it.

3 **Photograph 360°**
25–40 overlapping photos with a DSLR (SOP-3 settings) or phone (SOP-2), walking around the crown at two heights. The paper found 25 images to be the minimum for a complete model.

4 **Reconstruct**
Tools > Import from a capture device with the photo folder (COLMAP). Scale with ArUco in the 3D analysis tab settings.

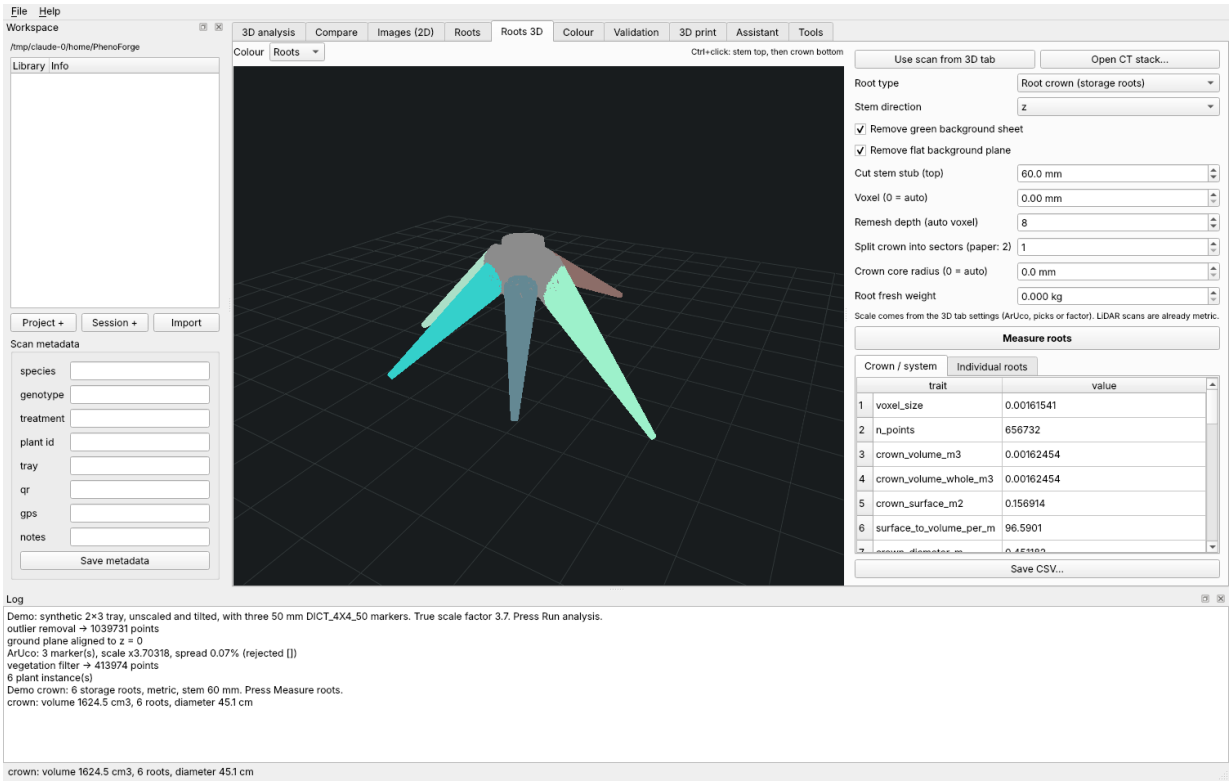
Measure in PhenoForge (Roots 3D tab)

1 **Load**
Use scan from 3D tab after import (or Help > Load demo root crown to practise).

2 **Orient**
Stem direction: the axis from the roots to the stem (e.g. *-y* if the crown lies on its side), or *two picked points*: Ctrl+click the stem top, then the crown bottom.

3 **Cut the stem**
Type the stub length to remove (the paper removes the stem before measuring).

4 **Measure**
Root type *Root crown*, then **Measure roots**. The scan is closed and filled with a voxel remesh (like Blender's remesh at octree depth 8), and each storage root is separated and coloured.



Roots 3D tab: demo crown with six storage roots separated.

Traits

Trait	Definition in PhenoForge	Paper equivalent
Crown volume	Filled voxel volume of the crown (optionally summed over sectors: 'split in half' = 2)	3D crown volume
Crown surface	Marching-cubes surface of the filled crown	3D crown surface area
Surface-to-volume	Surface ÷ volume	Same
Root density	Fresh weight ÷ crown volume (enter the weight)	Root density
Root number	3D components after removing the crown core	Root number
Root angle	Angle of each root (attachment → tip) from the vertical; mean reported	Root angle
Root length	Geodesic length from the crown core to the tip, per root	Root length
Crown diameter	Largest width across the axis	Crown diameter
Cylinder soil volume	π (diameter/2)² × crown depth	Cylinder soil volume
Compactness	Crown volume ÷ cylinder soil volume × 100	CRC compactness

Fibrous roots and CT

Washed cereal root systems and X-ray CT

- Washed roots: float the root system in water in a clear tray, or hang it, and photograph it as above; choose root type *Fibrous root system*.
- CT / MRI: **Open CT stack…** with the TIFF stack and the voxel size; roots are thresholded (Otsu) and the largest connected system is kept.
- Traits: total root length, number of tips and branch points, maximum depth and width, convex hull volume, solidity, root length by depth layer and the depth above which 50 % of the length lies (D50).

Accuracy on synthetic test objects

- Root crowns: crown volume within 2–3 %, all roots found, root angles within 4°, root lengths within ~10 %.
- Fibrous roots: total length about 9 % high (voxel staircase), tips within 1.
- Check against your own manual measurements (Validation tab) before relying on a trait.